

Question 1 (15 Marks)	Marks
(a) Factorise $8x^3 + 64y^3$.	2
(b) Simplify fully $\frac{2^{-x} \cdot 4^{2x+1}}{(2^{-x})^3}$.	2
(c) Find the equation of a line passing through the point of intersection of the straight lines $2x - 3y + 5 = 0$ and $3x + 4y + 8 = 0$ and the point $(2, 1)$.	3
(d) Differentiate with respect to x :	
(i) $3\pi^2 - 2$	1
(ii) $\frac{2x^3 - 5x}{x}$	2
(iii) $\frac{2x}{3 - 4x}$	2
(e) (i) State the natural domain and range for the function $y = \sqrt{3 - x}$.	2
(ii) Neatly sketch the function $y = \sqrt{3 - x}$.	1

Question 2 (15 Marks)	START A NEW PAGE	Marks
(a) Find the gradient of the tangent to the curve $y = \frac{1}{5 - 2x}$ at the point $(3, -1)$		2
(b) Solve for x :		
(i) $\tan x = 1$, where $0^\circ \leq x \leq 360^\circ$.		2
(ii) $\sqrt{3 - 2x} = x$		3
(iii) $\frac{18}{2x+1} - \frac{5}{x} = 1$		2

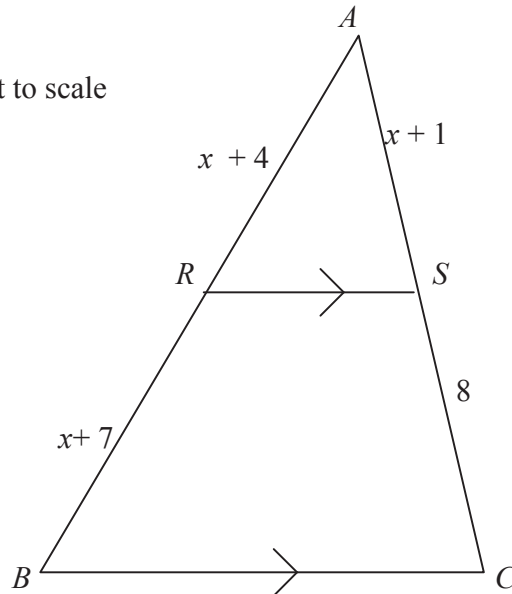
Question 2 continues overleaf.

Question 2 continued**Marks**

- (c) In the diagram below $RS \parallel BC$. Find the value of x , giving reasons.

2

Diagram not to scale



- (d) $ABCD$ is a quadrilateral such that $AB = 8$ units, $BC = 16$ units, $CD = 18$ units and $AD = 24$ units. Diagonal $AC = 12$ units. Draw a neat sketch on your answer page, clearly showing this information.

- (i) Prove that $\triangle ABC \parallel \triangle CAD$.

2

- (ii) Hence show that $AB \parallel DC$.

2**Question 3 (15 Marks) START A NEW PAGE****Marks**

- (a) Using a sketch, neatly indicate the region of the Cartesian plane for which the following inequalities hold true.

3

$$y \leq x, y \geq 0 \text{ and } y \leq \sqrt{9 - x^2}$$

- (b) Solve for x and y , given that $4^x \cdot 8^{-y} = 1$ and $25^x = 125 \times 5^y$.

3

- (c) Derive the equation of the locus of $P(x, y)$ such that its distance from $A(2, 4)$ is twice its distance from $B(3, 5)$.

3**Question 3 continues overleaf.**

Question 3 continued**Marks**

(d) $HF = 400\text{cm}$ and $GF = 200\text{cm}$. $\angle EFH = 15^\circ$.

(i) Find the size of $\angle HFG$.

1

(ii) Find the length of EH .

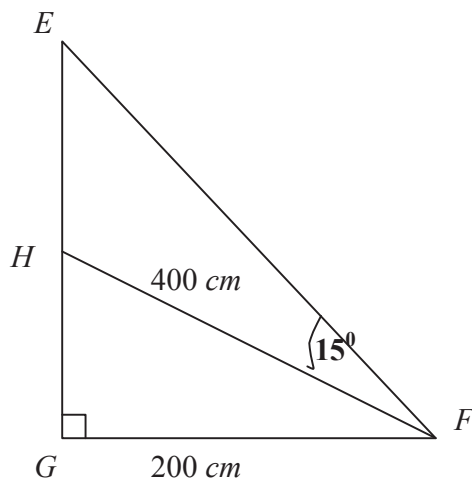
1

Diagram not to scale

(e) $A(1, 6)$ and $B(3, 2)$ are points on the curve $xy = 6$ and it is known that the tangent at B has slope $-\frac{2}{3}$.

(i) Find the equation of the tangent at B .

1

(ii) If this tangent meets the x, y axes at E, F respectively, prove that B is the midpoint of EF .

3**Question 4 (15 Marks)****START A NEW PAGE****Marks**

(a) The line $x + y = 3\sqrt{2}$ is a tangent to a circle with centre $(0, 0)$.
Find the radius of this circle.

2

(b) If $f(x)$ and $g(x)$ are odd functions and $h(x) = f(x).g(x)$, prove that $h(x)$ is an even function.

2

(c) The lines $3x - y + 3 = 0$, $y = 3x - 6$, $y = x - 4$ and $5x + 3y = 15$ form a quadrilateral. Describe the type of quadrilateral formed, giving clear reasons.

3**Question 4 continued overleaf.**

Question 4 continued.**Marks**

(d) Solve for x :

(i) $|3x - 2| < |3 + 2x|$ **3**

(ii) $\cos(180^\circ - x) = \frac{1}{2}; 0^\circ \leq x \leq 360^\circ$ **2**

(e) Show, by the method of First principles, that the derivative of $y = \sqrt{2x}$ **3**
is given by $\frac{dy}{dx} = \frac{1}{\sqrt{2x}}$.

~END OF TEST ~

Year 11 Preliminary Half Yearly Maths (2U) Examinations 2007 – SOLUTIONS

Question 1

(a) $8x^3 + 64y^3 = (2x + 4y)(4x^2 - 8xy + 16y^2)$

→ 2 Marks

(b)
$$\frac{2^{-x} \cdot 4^{2x+1}}{(2^{-x})^3} = \frac{2^{-x} \cdot 2^{4x+2}}{2^{-3x}}$$
$$= 2^{6x+2}$$

→ 2 Marks

(c) New line must satisfy $2x - 3y + 5 + k(3x + 4y + 8) = 0$ at $(2, 1)$.

→ 1 Mark

$$\therefore 4 - 3 + 5 + k(6 + 4 + 8) = 0 \rightarrow k = -\frac{1}{3}$$

→ 1 Mark for k

$$\therefore \text{eqn of new line is: } 3x - 13y + 7 = 0$$

→ 1 Mark for equation.

(d) (i) $\frac{d}{dx}(3\pi^2 - 2) = 0$

→ 1 Mark

(ii)
$$\frac{d}{dx}\left(\frac{2x^3 - 5x}{x}\right) = \frac{d}{dx}(2x^2 - 5), x \neq 0$$
$$= 4x$$

→ 1 Mark for simplification

→ 1 Mark correct answer

(iii)
$$\frac{d}{dx}\left(\frac{2x}{3-4x}\right) = \frac{(3-4x)(2) - (2x)(-4)}{(3-4x)^2}$$
$$= \frac{6}{(3-4x)^2}$$

→ 1 Mark correct quotient rule

→ 1 Mark correct answer

Alternative solution

$$\frac{d}{dx}(2x)(3-4x)^{-1} = (2x) \cdot -1(3-4x)^{-2} \cdot -4 + (3-4x)^{-1} \cdot 2$$
$$= \frac{8x}{(3-4x)^2} + \frac{2}{(3-4x)}$$
$$= \frac{6}{(3-4x)^2}$$

→ correct use of rule
1 Mark

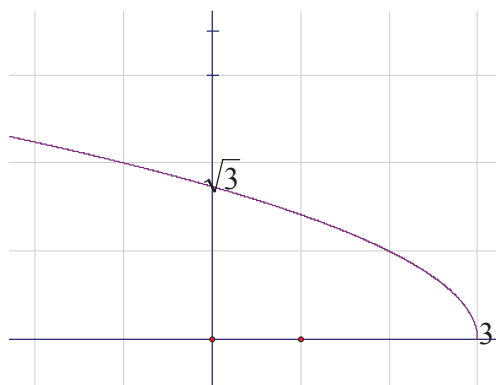
→ 1 Mark correct answer

(e) (i) Domain: $\{x : x \leq 3\}$; Range: $\{y : y \geq 0\}$

→ 1 Mark each

(ii)

→ 1 Mark



Question 2

(a) $\frac{d}{dx}\left(\frac{1}{5-2x}\right) = \frac{2}{(5-2x)^2}$

$\therefore$ gradient at $(3, -1)$ is $m = 2$

→ 1 Mark correct differentiation

→ 1 Mark

(b) (i) $\tan x = 1$
 $x = 45^\circ$ or 225°

→ 1 Mark each

(ii) $\sqrt{3-2x} = x$ [Restriction : $0 \leq x \leq 1.5$]
 $\therefore 3-2x = x^2 \rightarrow x^2 + 2x - 3 = 0$
 $\therefore (x+3)(x-1) = 0$
 $\therefore x = -3$ or $x = 1$ $\therefore$ **only solution is $x = 1$**

→ 1 Mark restriction

→ 1 Mark

→ 1 Mark answer with justification

(iii) $\frac{18}{2x+1} - \frac{5}{x} = 1$ [$x \neq 0$ or $x \neq -\frac{1}{2}$]
 $\therefore 18x - 5(2x+1) = x(2x+1)$
 $\therefore 2x^2 - 7x + 5 = 0$
 $\therefore (2x-5)(x-1) = 0$
 $\therefore x = 1$ or $x = \frac{5}{2}$

→ 1 Mark

→ 1 Mark

(c) $\frac{x+4}{x+7} = \frac{x+1}{8}$

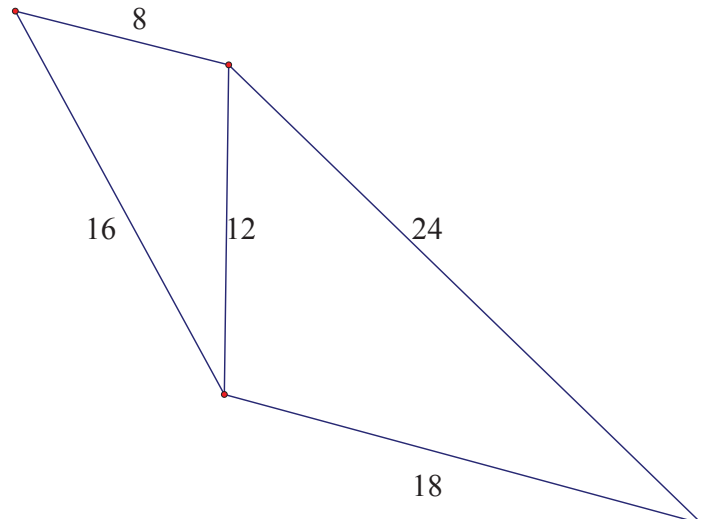
→ 1 Mark correct ratio with reason

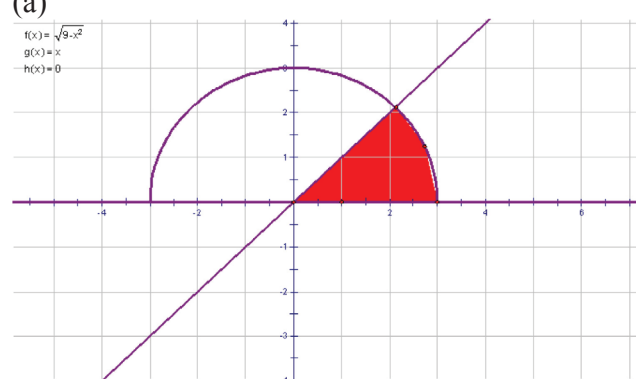
(A line parallel to one side of a triangle divides the other 2 sides in the same ratio)

$\therefore 8x + 32 = x^2 + 8x + 7$
 $\therefore x^2 = 25 \rightarrow x = 5$ ($x > 0$)

→ 1 Mark answer with justification

(d) Next page

 (d) (i) In $\triangle ABC$ and $\triangle ADC$ $\frac{AB}{AC} = \frac{8}{12} = \frac{2}{3}$, $\frac{AC}{DC} = \frac{12}{18} = \frac{2}{3}$, $\frac{BC}{AD} = \frac{16}{24} = \frac{2}{3}$ $\therefore \triangle ABC \parallel \triangle ADC$ (3 corresponding sides in proportion) (ii) $\angle BAD = \angle ACD$ (Corresponding angles of similar triangles are equal) $\therefore AB \parallel DC$ (Alternate angles equal only on parallel lines)	→ 1 Mark correct sides in ratio → correct reasoning 1 Mark → 1 Mark → 1 Mark
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Question 3 (a)  (b) $4^x \cdot 8^{-y} = 1 \Rightarrow 2x - 3y = 0 \dots (A)$ $25^x = 125 \times 5^y \Rightarrow 2x = 3 + y \dots (B)$ Subst. (B) into (A) $\therefore 3 - 2y = 0 \therefore y = \frac{3}{2}$ $\therefore 2x = 4.5 \therefore x = \frac{9}{4}$	→ 1 Mark $y = \sqrt{9 - x^2}$ → 1/2 mark each for $y = x$ and $y = 0$ → 1 Mark correct region → 1 Mark for (A) → 1 Mark for (B) → 1 Mark for answer x and y
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Question 3

(c) $PA = 2PB \rightarrow PA^2 = 4PB^2$

$$\therefore (x-2)^2 + (y-4)^2 = 4[(x-3)^2 + (y-5)^2]$$

$$\therefore x^2 - 4x + 4 + y^2 - 8y + 16 = 4x^2 - 24x + 36 + 4y^2 - 40y + 100$$

$$\therefore 3x^2 + 3y^2 - 20x - 32y + 116 = 0 \rightarrow \text{eqn of circle.}$$

→ 1 Mark

→ 1 Mark

→ 1 Mark

(d) (i) $\cos \angle HFG = \frac{200}{400} \rightarrow \angle HFG = 60^\circ$

→ 1 Mark

(ii) $\angle HEF = 15^\circ$ (Angle sum of $\triangle EGF$)

$$\therefore EH = HF \text{ (Equal sides opposite equal angles)}$$

$$\therefore EH = 400 \text{ cm}$$

→ ½ Mark

→ ½ Mark

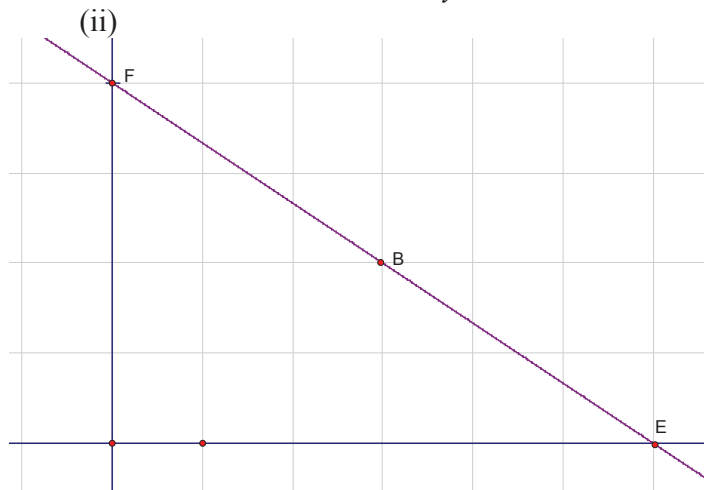
(e) (i) gradient = $-\frac{2}{3}$ (given)

$$\therefore \text{eqn. of tangent is: } y - 2 = -\frac{2}{3}(x - 3)$$

$$\text{i.e. } 2x + 3y - 12 = 0$$

→ 1 Mark

(eqn. can be of any form.)



$$E(6, 0) \text{ and } F(0, 4) \therefore \text{midpoint} = \left(\frac{6+0}{2}, \frac{0+4}{2} \right) = (3, 2)$$

$\therefore B$ is the midpoint of EF .

→ Point E 1 Mark

→ point F 1 Mark

→ Showing B is the midpoint 1 Mark

Question 4

- (a) If line is a tangent then the radius = perp. Distance of the line from the centre of the circle.

$$\therefore \text{Radius} = \left| \frac{0+0-3\sqrt{2}}{\sqrt{2}} \right| = 3 \text{ units}$$

- (b) If $f(x)$ and $g(x)$ are odd functions then by definition $f(-x) = -f(x)$ and $g(-x) = -g(x)$

$$\begin{aligned} \therefore h(-x) &= f(-x) \times g(-x) \\ &= -f(x) \times -g(x) \\ &= f(x) \times g(x) = h(x) \end{aligned}$$

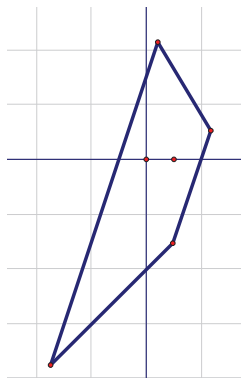
$\therefore$ since $h(-x) = h(x)$; **$h(x)$ is an EVEN function**

- (c) $L_1: 3x - y + 3 = 0 \rightarrow$ gradient $m_1 = 3$

$$L_2: y = 3x - 6 \rightarrow \text{gradient } m_2 = 3$$

$$L_3: y = x - 4 \rightarrow \text{gradient } m_3 = 1$$

$$L_4: 5x + 3y = 15 \rightarrow m_4 = -\frac{5}{3}$$



$\therefore$ since $m_1 = m_2$ we have $L_1 \parallel L_2$
 $\therefore$ quadrilateral formed is a trapezium (one pair of opposite sides parallel)

- (d) (i) $|3x - 2| < |3 + 2x|$

$$\therefore 9x^2 - 12x + 4 < 9 + 12x + 4x^2 \text{ (by squaring both sides)}$$

$$\therefore 5x^2 - 24x - 5 < 0$$

$$\therefore (5x + 1)(x - 5) < 0 \rightarrow -\frac{1}{5} < x < 5$$

(ii) $\cos(180^\circ - x) = \frac{1}{2}$

$$\cos x = -\frac{1}{2} \text{ (by supplementary angles)}$$

$$\therefore x = 120^\circ \text{ or } x = 240^\circ$$

$\rightarrow$ 1 Mark correct use perp. Distance equation
 $\rightarrow$ 1 Mark correct answer

$\rightarrow$ 1 Mark

$\rightarrow$ 1 Mark

$\rightarrow$ 1 Mark

$\rightarrow$ 1 Mark

$\rightarrow$ 1 Mark

$\rightarrow$ 1 Mark

$\rightarrow$ 1 Mark Correct solution

$\rightarrow$ 1 Mark

$\rightarrow$ 1 Mark

Question 4

(d) $y = \sqrt{2x}$

$$\therefore \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{(y + \Delta y) - y}{\Delta x} \text{ by First principles of diff.}$$

$$\therefore \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\sqrt{2(x + \Delta x)} - \sqrt{2x}}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{2(x + \Delta x)} - \sqrt{2x}}{\Delta x} \times \frac{\sqrt{2(x + \Delta x)} + \sqrt{2x}}{\sqrt{2(x + \Delta x)} + \sqrt{2x}}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{2x + 2\Delta x - 2x}{\Delta x (\sqrt{2x + 2\Delta x} + \sqrt{2x})}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{2\Delta x}{\Delta x (\sqrt{2x + 2\Delta x} + \sqrt{2x})}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{2}{(\sqrt{2x + 2\Delta x} + \sqrt{2x})}$$

$$= \frac{2}{(\sqrt{2x} + \sqrt{2x})} \text{ as } \Delta x \rightarrow 0$$

$$= \frac{2}{2\sqrt{2x}} = \frac{1}{\sqrt{2x}} \text{ as required.}$$

→ correct use of first principles **1** Mark

→ **1** Mark multiply by the conjugate of numerator

→ **1** Mark correct simplification to result to answer